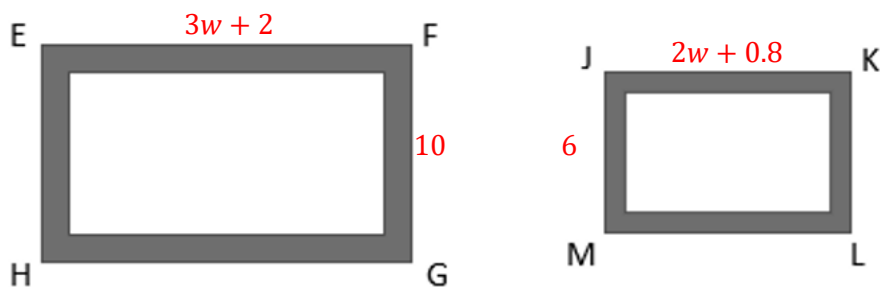


Geometry
Unit 6
Lesson 12 Practice

Name Answer Key
Date _____
Period _____

Tamika is custom building frames for a set of family photos. She has designed two similar rectangular frames, $EFGH$ and $JKLM$. The diagram shows the two different sized frames, where $JM = 6$ inches (in.), $EF = (3w + 2)$ in., $FG = 10$ in., and $JK = (2w + 0.8)$ in.



1. What is the value of w ?

$$\frac{2w+0.8}{3w+2} = \frac{6}{10}$$

$$10(2w + 0.8) = 6(3w + 2)$$

$$2w = 4$$

$$20w + 8 = 18w + 12$$

$$w = 2$$

2. What is the area of $EFGH$? Include units.

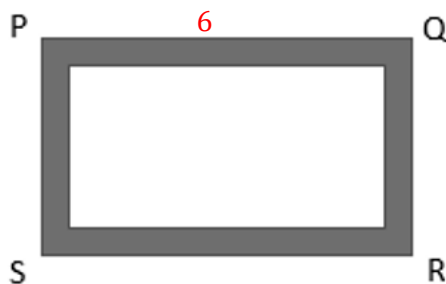
$$A = EF \cdot FG = [3(2) + 2] \cdot 10 = 80 \text{ in}^2$$

3. What is the perimeter of $JKLM$? Include units.

$$JK = 2(2) + 0.8 = 4.8 \text{ in}$$

$$P = 2 \cdot 4.8 + 2 \cdot 6 = 21.6 \text{ in}$$

Tamika created a third similar rectangular frame, $PQRS$, shown below, where $PQ = 6$ in.



4. What is the length of $\overline{SP}$? Include units.

$$SP = 7.5 \text{ in}$$

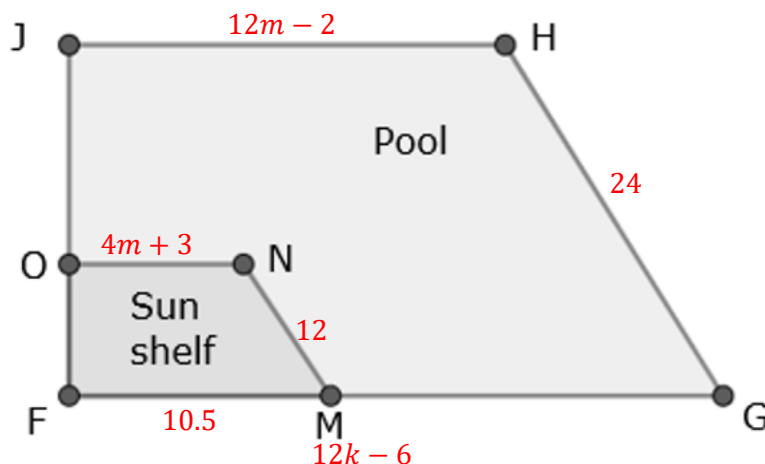
$$\frac{6}{8} = \frac{SP}{10}$$

$$SP = \frac{(10)(6)}{8} = 7.5 \text{ in}$$

5. What is the perimeter, in inches, of $PQRS$?

$$P = 2 \cdot 6 + 2 \cdot 7.5 = 27 \text{ in}$$

Shelley and her family are having a pool built in the backyard. The pool, $FGHJ$, is shaped like a trapezoid, with a smaller similar trapezoid, $FMNO$, for a sun shelf. A diagram of the pool is shown, where $FG = (12k - 6)$ feet (ft), $FM = 10.5$ ft, $NM = 12$ ft, $HG = 24$ ft, $HJ = (12m - 2)$ ft, and $NO = (4m + 3)$ ft.



6. Determine the value of each variable:

$$k = \underline{2.25} \quad \frac{10.5}{12k-6} = \frac{12}{24} \quad 144k = 324 \quad \left| \quad \frac{4m+3}{12m-2} = \frac{12}{24} \quad 2m = 4 \right.$$

$$m = \underline{2} \quad (24)(10.5) = 12(12k - 6) \quad k = 2.25 \quad \left| \quad 4m + 3 = \frac{1}{2}(12m - 2) \quad m = 2 \right.$$

$$252 = 144k - 72 \quad \left| \quad 4m + 3 = 6m - 1 \right.$$

7. Determine the value of each side:

$$HJ = \underline{22} \quad HJ = 12m - 2 = 12(2) - 2 = 22$$

$$FG = \underline{21} \quad FG = 12k - 6 = 12(2.25) - 6 = 21$$

8. If $HJ = JF$, what is the perimeter, in feet, of the pool?

$$P = HJ + JF + FG + HG = 22 + 22 + 21 + 24 = \boxed{89 \text{ ft}}$$

Some measurements of angles of $FGHJ$ and $FMNO$ are given.

$$m\angle FMN = (15y + 10)^\circ, m\angle FGH = (18y - 2)^\circ, m\angle OFM = 90^\circ,$$

$$m\angle FJH = (5w + 15)^\circ, m\angle FON = (15w - 15)^\circ$$

9. Determine the value of each variable:

$$y = \underline{4} \quad 15y + 10 = 18y - 2 \quad 5w + 15 = 15w - 15$$

$$w = \underline{3} \quad 3y = 12 \quad 10w = 30$$

$$y = 4 \quad w = 3$$

10. What is the measure of $\angle ONM$?

$$m\angle OFM + m\angle FON + m\angle ONM + m\angle FMN = 360^\circ$$

$$\boxed{m\angle ONM = 170^\circ}$$

$$90^\circ + [(15)(3) - 15]^\circ + m\angle ONM + [(15)(4) + 10]^\circ = 360^\circ$$

$$90^\circ + 30^\circ + m\angle ONM + 70^\circ = 360^\circ$$